

Definite Integrals

Definition

(i) If $a \in \text{Dom}(f)$, then $\int_a^a f(x) dx = 0$

(ii) If f is integrable on $[a, b]$, then $\int_a^b f(x) dx = -\int_b^a f(x) dx$

Theorem

If f and g are integrable on $[a, b]$ and k is a constant, then kf , $f + g$ and $f - g$ are integrable on $[a, b]$ and then

$$(i) \int_a^b kf(x) dx = k \int_a^b f(x) dx$$

$$(ii) \int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$(iii) \int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

In general,

$$\int_a^b [f_1(x) \pm f_2(x) \pm \dots \pm f_n(x)] dx = \int_a^b f_1(x) dx \pm \int_a^b f_2(x) dx \pm \dots \pm \int_a^b f_n(x) dx$$

Some properties of definite integrals can be motivated. For example, if f is continuous and nonnegative on the interval $[a, b]$, and if c is a point between a and b , then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Some important properties for evaluating integrals

$$(1) \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$(2) \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f \text{ is even} \\ 0 & \text{if } f \text{ is odd} \end{cases}$$

$$(3) \int_0^{na} f(x) dx = n \int_0^a f(x) dx$$

Solved Problems

1. Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$

Solution: Let $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$ (1)

Then $I = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \{\cos(\pi - x)\}^2} dx = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$ (2)

Adding equations (1) and (2), we get

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = 2\pi \int_0^{\pi/2} \frac{\sin x}{1 + \cos^2 x} dx \Rightarrow I = \pi \int_0^{\pi/2} \frac{\sin x}{1 + \cos^2 x} dx$$
 (3)

Let $\cos x = u$. Then $-\sin x dx = du$. Also, if $x = 0$ then $u = 1$ and if $x = \pi/2$ then $u = 0$.

So that $I = -\pi \int_1^0 \frac{1}{1 + u^2} du = \pi [\tan^{-1} u]_0^1 = \pi(\pi/4 - 0) = \frac{\pi^2}{4}$.

2. Evaluate $\int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx$;

Solution: Let $I = \int_0^{\pi} \frac{x \tan x}{\sec x + \cos x} dx = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \pi^2/4$.

3. Evaluate $\int_0^{\pi} \frac{x}{1 + \sin x} dx$;

Solution: Let $I = \int_0^{\pi} \frac{x}{1 + \sin x} dx$ (1)

Then $I = \int_0^{\pi} \frac{\pi - x}{1 + \sin(\pi - x)} dx = \int_0^{\pi} \frac{\pi - x}{1 + \sin x} dx$ (2)

Adding equations (1) and (2), we get

$$2I = \pi \int_0^{\pi} \frac{1}{1 + \sin x} dx = 2\pi \int_0^{\pi/2} \frac{1}{1 + \sin(\pi/2 - x)} dx$$

$$\Rightarrow I = \pi \int_0^{\pi/2} \frac{1}{1 + \cos x} dx = \pi \int_0^{\pi/2} \frac{1}{2 \cos^2(x/2)} dx = \frac{\pi}{2} \int_0^{\pi/2} \sec^2(x/2) dx$$

$$= \frac{\pi}{2} \times 2 [\tan \frac{x}{2}]_0^{\pi/2} = \pi \tan \frac{\pi}{4} = \pi.$$

4. Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \sin x} dx$

Solution: Let $I = \int_0^{\pi} \frac{x \sin x}{1 + \sin x} dx$ (1)

Then $I = \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + \sin(\pi - x)} dx = \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + \sin x} dx$ (2)

Adding equations (1) and (2), we get

$$\begin{aligned} 2I &= \pi \int_0^{\pi} \frac{\sin x}{1 + \sin x} dx = 2\pi \int_0^{\pi/2} \frac{\sin x}{1 + \sin x} dx \\ \Rightarrow I &= \pi \int_0^{\pi/2} \frac{\sin\{(\pi/2) - x\}}{1 + \sin\{(\pi/2) - x\}} dx = \pi \int_0^{\pi/2} \frac{\cos x}{1 + \cos x} dx = \pi \int_0^{\pi/2} \frac{(1 + \cos x) - 1}{1 + \cos x} dx \\ &= \pi \frac{\pi}{2} - \frac{\pi}{2} \times 2 [\tan \frac{x}{2}]_0^{\pi/2} = \frac{\pi^2}{2} - \pi \tan \frac{\pi}{4} = \pi \left(\frac{\pi}{2} - 1 \right). \end{aligned}$$

5. Evaluate $\int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx$

Solution: Let $I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx$ (1)

Then $I = \int_0^{\pi/2} \frac{(\pi/2) - x}{\sin((\pi/2) - x) + \cos((\pi/2) - x)} dx$

$$\Rightarrow I = \int_0^{\pi/2} \frac{(\pi/2) - x}{\cos x + \sin x} dx$$
 (2)

Therefore, adding (1) and (2), we get

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{\sin x + \cos x} dx = \frac{\pi}{2} \int_0^{\pi/2} \frac{\sec^2(x/2)}{1 - \tan^2(x/2) + 2 \tan(x/2)} dx$$
 (3)

Let $\tan(x/2) = y \Rightarrow (1/2) \sec^2(x/2) dx = dy \Rightarrow \sec^2(x/2) dx = 2dy$

Here, $x = 0 \Rightarrow y = 0$ and $x = \pi/2 \Rightarrow y = 1$

$$\begin{aligned} \text{Then } 2I &= \frac{\pi}{2} \int_0^1 \frac{2dy}{1 - y^2 + 2y} = \pi \int_0^1 \frac{dy}{(\sqrt{2})^2 - (y - 1)^2} = \pi \times \frac{1}{2\sqrt{2}} \left[\log \frac{\sqrt{2} + y - 1}{\sqrt{2} - y + 1} \right]_0^1 \\ &= \pi \times \frac{1}{2\sqrt{2}} \left[\log \frac{\sqrt{2}}{\sqrt{2}} - \log \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \right] = \pi \times \frac{1}{2\sqrt{2}} \log \frac{\sqrt{2} + 1}{\sqrt{2} - 1} \\ &= \pi \times \frac{1}{2\sqrt{2}} \log \frac{(\sqrt{2} + 1)^2}{2 - 1} = \pi \times \frac{1}{\sqrt{2}} \log(\sqrt{2} + 1) \end{aligned}$$

Thus, $I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx = \frac{\pi}{2\sqrt{2}} \log(\sqrt{2} + 1)$

5. Evaluate $\int_0^{\pi/2} \log(\cos x) dx$

Solution: Let $I = \int_0^{\pi/2} \log(\cos x) dx$ (1)

Then $I = \int_0^{\pi/2} \log\{\cos(\frac{\pi}{2} - x)\} dx = \int_0^{\pi/2} \log \sin x dx$ (2)

Adding equations (1) and (2), we get

$$\begin{aligned} 2I &= \int_0^{\pi/2} \log(\cos x) dx + \int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\sin x \cos x) dx = \int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right) dx \\ &= \int_0^{\pi/2} \log(\sin 2x) dx - \int_0^{\pi/2} \log(2) dx = I_1 - \frac{\pi}{2} \log 2 \end{aligned}$$

For I_1 , let $2x = z$. Then $2dx = dz \Rightarrow dx = \frac{1}{2} dz$

So, $I_1 = \int_0^{\pi/2} \log(\sin 2x) dx = \int_0^{\pi} \log(\sin z) \frac{1}{2} dz = \frac{1}{2} \times 2 \int_0^{\pi/2} \log(\sin x) dx = I$

Hence, $2I = I - \frac{\pi}{2} \log 2 \Rightarrow I = -\frac{\pi}{2} \log 2 = \frac{\pi}{2} \log \frac{1}{2}$.

6. Evaluate $\int_3^{29} \frac{(x-2)^{2/3}}{(x-2)^{2/3} + 3} dx$.

Solution: Let $(x-2)^{2/3} = y^2$. Then $x-2 = y^3 \Rightarrow dx = 3y^2 dy$

When $x = 3$ then $y = 1$ and when $x = 29$ then $y = 3$

Therefore, we can write

$$\begin{aligned} \int_3^{29} \frac{(x-2)^{2/3}}{(x-2)^{2/3} + 3} dx &= \int_1^3 \left(1 - \frac{3}{(x-2)^{2/3} + 3}\right) dx = \int_1^3 \left(1 - \frac{3}{y^2 + 3}\right) 3y^2 dy \\ &= 3 \int_1^3 \left(y^2 - \frac{3y^2}{y^2 + 3}\right) dy = 3 \int_1^3 y^2 dy - 9 \int_1^3 \left(1 - \frac{3}{y^2 + 3}\right) dy \\ &= 3 \left[\frac{y^3}{3}\right]_1^3 - 9 \left[y - \frac{3}{\sqrt{3}} \tan^{-1} \frac{y}{\sqrt{3}}\right]_1^3 = 26 - 9 \left(2 - \sqrt{3} \tan^{-1} \sqrt{3} + \sqrt{3} \tan^{-1} \frac{1}{\sqrt{3}}\right) \\ &= 26 - 9 \left(2 - \sqrt{3} \frac{\pi}{3} + \sqrt{3} \frac{\pi}{6}\right) = 8 + \frac{3\sqrt{3}}{2} \pi. \end{aligned}$$

7. Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2}$

Solution: We can write

$$\int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2} = \int_0^{\frac{\pi}{2}} \frac{\sec^4 x dx}{(a^2 + b^2 \tan^2 x)^2}$$

Let $b \tan x = a \tan y \Rightarrow \sec^2 x dx = (a/b) \sec^2 y dy$

Here, $x = 0 \Rightarrow y = 0$ and $x = \pi/2 \Rightarrow y = \pi/2$

$$\begin{aligned} \text{Therefore, we get } \int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2} &= \int_0^{\frac{\pi}{2}} \frac{(\sec^2 x)(\sec^2 x) dx}{(a^2 + b^2 \tan^2 x)^2} \\ &= \int_0^{\frac{\pi}{2}} \frac{(1 + \frac{a^2}{b^2} \tan^2 y)(a/b) \sec^2 y dy}{(a^2 + a^2 \tan^2 y)^2} = \int_0^{\frac{\pi}{2}} \frac{(b^2 + a^2 \tan^2 y)}{a^3 b^3 \sec^2 y} dy = \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} (b^2 \cos^2 y + a^2 \sin^2 y) dy \end{aligned}$$

$$\begin{aligned} \text{Let } I &= \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} (b^2 \cos^2 y + a^2 \sin^2 y) dy \\ &= \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} \{b^2 \cos^2(\frac{\pi}{2} - y) + a^2 \sin^2(\frac{\pi}{2} - y)\} dy = \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} (b^2 \sin^2 y + a^2 \cos^2 y) dy \end{aligned}$$

$$\text{So that } 2I = \frac{1}{a^3 b^3} \int_0^{\frac{\pi}{2}} (a^2 + b^2) dy = \frac{a^2 + b^2}{a^3 b^3} \frac{\pi}{2}$$

$$\text{Thus, we get } \int_0^{\frac{\pi}{2}} \frac{dx}{(a^2 \cos^2 x + b^2 \sin^2 x)^2} = \frac{a^2 + b^2}{a^3 b^3} \frac{\pi}{4}$$

8. Evaluate $\int_0^{\infty} \frac{x dx}{(1+x)(1+x^2)}$

Solution: Let $x = \tan y \Rightarrow dx = \sec^2 y dy$

Here, $x = 0 \Rightarrow y = 0$ and $x = \infty \Rightarrow y = \pi/2$

$$\begin{aligned} \text{Therefore, } \int_0^{\infty} \frac{x dx}{(1+x)(1+x^2)} &= \int_0^{\pi/2} \frac{\tan y \sec^2 y}{(1+\tan y) \sec^2 y} dy \\ &= \int_0^{\pi/2} \frac{\tan y}{1+\tan y} dy = \int_0^{\pi/2} \frac{\sin y}{\sin y + \cos y} dy = I, \text{ say} \end{aligned}$$

$$\text{Then } I = \int_0^{\pi/2} \frac{\sin((\pi/2) - y)}{\sin((\pi/2) - y) + \cos((\pi/2) - y)} dy = \int_0^{\pi/2} \frac{\cos y}{\cos y + \sin y} dy$$

Thus, $2I = \int_0^{\pi/2} dy = \pi/2 \Rightarrow I = \int_0^{\infty} \frac{x dx}{(1+x)(1+x^2)} = \int_0^{\pi/2} \frac{\sin y}{\sin y + \cos y} dy = \pi/4$

9. Evaluate $\int_0^{\infty} \frac{\log(x + \frac{1}{x})}{1+x^2} dx$

Solution: Let $x = \tan y \Rightarrow dx = \sec^2 y dy$

Here, $x = 0 \Rightarrow y = 0$ and $x = \infty \Rightarrow y = \pi/2$

$$\begin{aligned} \text{Therefore, } I &= \int_0^{\infty} \frac{\log(x + \frac{1}{x})}{1+x^2} dx = \int_0^{\pi/2} \frac{\log(\tan y + \cot y) \sec^2 y}{1+\tan^2 y} dy \\ &= \int_0^{\pi/2} \log \left[\frac{1+\tan^2 y}{\tan y} \right] dy = \int_0^{\pi/2} \log \left[\frac{\sec^2 y}{\tan y} \right] dy = \int_0^{\pi/2} \log \left[\frac{1}{\sin y \cos y} \right] dy \\ &= \int_0^{\pi/2} \log \left[\frac{2}{2 \sin y \cos y} \right] dy = \int_0^{\pi/2} \log 2 dy - \int_0^{\pi/2} \log(\sin 2y) dy = \frac{\pi}{2} \log 2 - I_1, \end{aligned}$$

where $I_1 = \int_0^{\pi/2} \log(\sin 2y) dy$

Let $z = 2y \Rightarrow dz = 2 dy \Rightarrow dy = (1/2) dz$

Here, $y = 0 \Rightarrow z = 0$ and $y = \pi/2 \Rightarrow z = \pi$

$$\begin{aligned} \text{Therefore, } I_1 &= \int_0^{\pi/2} \log(\sin 2y) dy = \frac{1}{2} \int_0^{\pi} \log(\sin z) dz \\ &= \int_0^{\pi/2} \log(\sin z) dz = \frac{\pi}{2} \log \frac{1}{2} \end{aligned}$$

Thus, $I = \int_0^{\infty} \frac{\log(x + \frac{1}{x})}{1+x^2} dx = \frac{\pi}{2} \log 2 - \frac{\pi}{2} \log \frac{1}{2} = 2 \times \frac{\pi}{2} \log 2 = \pi \log 2$

Exercises

Evaluate the following definite integrals:

$$1. \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$$

$$2. \int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan x} dx$$

$$3. \int_0^{\frac{\pi}{2}} \log(\sin x) dx$$

$$4. \int_0^{\pi} \log(1 + \cos x) dx$$

$$5. \int_0^{\frac{\pi}{2}} \frac{\sin^{2m-1} \theta \cos^{2n-1} \theta}{(a \sin^2 \theta + b \cos^2 \theta)^{m+n}} d\theta$$

$$6. \int_0^{\infty} \frac{dx}{\sqrt{x}(x+1)}$$

$$7. \int_0^1 \frac{\log(1+x)}{1+x^2} dx$$

$$8. \int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \sin x \cos x} dx$$

$$9. \int_0^1 x^2 \sqrt{4-x^2} dx$$

$$10. \int_0^1 \frac{\log x}{\sqrt{1-x^2}} dx$$